Linear Algebra and Its Applications

Section 2.8: Subspaces of Rn

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.8 Subspaces of Rn

A subspace contains zero and is closed under *every* vector sum and *every* scalar multiple of its members. A single failure disproves subspace status. A span is always a subspace. For an $m \times n$ matrix,

$$\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \operatorname{Nul} A = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

Column space concerns attainable *outputs*; null space concerns *inputs* with zero output. A basis is an independent spanning set. Row reduction preserves null space and dependence relations between columns, but generally changes the column space itself. To construct a column-space basis, take the *original* columns in the pivot positions. To construct a null-space basis, solve $Ax = 0$, choose the free variables as independent parameters, and read off their coefficient vectors.

Practice problem 2.8.P1: Membership in two very different spaces

For the supplied A and $u = (-7, 3, 2)^T$,

$$Au = \begin{bmatrix} -7 - 3 + 10 \\ -14 + 14 \\ 21 - 15 - 6 \end{bmatrix} = 0.$$

Thus $u \in \operatorname{Nul} A$. It is not in $\operatorname{Col} A$. Indeed row 3 of A equals five times row 1 minus four times row 2. Every possible output $b = Ax$ must consequently obey $b_3 = 5b_1 - 4b_2$. For u, that would require $2 = 5(-7) - 4(3) = -47$, which is false. Thus the same vector can belong to the null space but not the column space, even when these spaces happen to share the same ambient dimension.

Practice problem 2.8.P2: Read a shift matrix

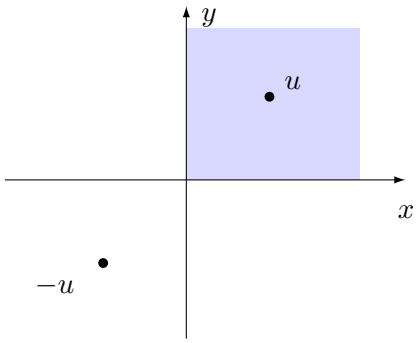
Here $A(x_1, x_2, x_3)^T = (x_2, x_3, 0)^T$. Thus $e_1 = (1, 0, 0)^T$ is a nonzero null vector. The second column of A is e_1, so e_1 is also a nonzero vector in $\operatorname{Col} A$. Alternatively e_2 belongs to the column space as column 3. It is legitimate for a vector to lie in both spaces in this example.

Practice problem 2.8.P3: The spaces of an invertible matrix

If A is invertible, every $b \in \mathbb{R}^n$ equals $A(A^{-1}b)$, so $\operatorname{Col} A = \mathbb{R}^n$. Also $Ax = 0$ implies $x = A^{-1}0 = 0$, so $\operatorname{Nul} A = \{0\}$. The zero subspace contains one vector; it is not an empty set.

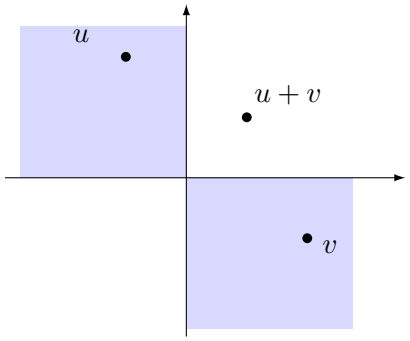
Exercise 2.8.1: A quadrant fails negative scaling

The shaded set is the first quadrant, including its axes. The vector $u = (1, 1)$ belongs to it, but $-u = (-1, -1)$ does not. Since -1 is an allowed scalar, closure under scalar multiplication fails. The following is a finite-window sketch; the quadrant continues outward.



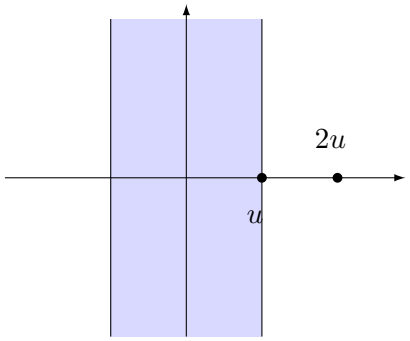
Exercise 2.8.2: Two allowed quadrants fail addition

The set consists of quadrants II and IV, with boundary axes. Take $u = (-1, 2)$ in quadrant II and $v = (2, -1)$ in quadrant IV. Their sum is $(1, 1)$ in quadrant I, outside the set. Thus addition is not closed.



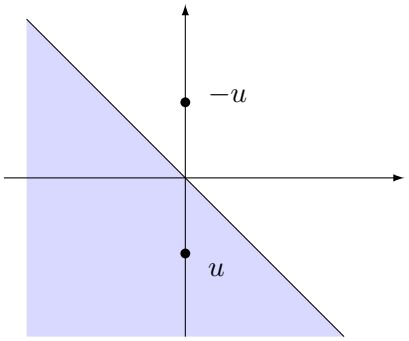
Exercise 2.8.3: A strip fails large scaling

Let the positive vertical boundary have coordinate $x = a > 0$. The point $u = (a, 0)$ lies on that included boundary, but $2u = (2a, 0)$ lies to its right, outside the strip. Hence closure under scalar multiplication fails. The finite upper and lower ends in the sketch are viewing limits, not extra boundaries.



Exercise 2.8.4: One side of a line is not a vector space

The shaded half-plane includes the region below a line through the origin. Take a point on the negative vertical axis, such as $u = (0, -1)$. It belongs to the shaded region, but $-u = (0, 1)$ lies on the unshaded side. Thus multiplying by -1 leaves the set, so it is not a subspace.



Exercise 2.8.5: Exhibit the required linear combination

Compute

$$v_1 + 2v_2 = (1, 3, -4)^T + 2(-2, -3, 7)^T = (-3, -3, 10)^T = w.$$

Thus $w \in \text{Span}\{v_1, v_2\}$. Giving coefficients that reproduce every coordinate proves membership directly; merely noticing that all vectors have three entries would not prove it.

Exercise 2.8.6: A consistency obstruction in four dimensions

Form $A = [v_1\ v_2\ v_3]$. Its four rows obey the relation $-11\,\text{row}_1 A + 5\,\text{row}_2 A + 10\,\text{row}_3 A + 2\,\text{row}_4 A = 0$. This identity is checked by its three entries: $-11 - 15 + 20 + 6 = 0$, $-44 - 20 + 50 + 14 = 0$, and $-55 - 15 + 60 + 10 = 0$. If $Ac = u$ had a solution, the same combination of its right-hand sides would be zero. Instead it equals $-11(-1) + 5(-7) + 10(-1) + 2(2) = -30$. This contradiction proves $u \notin \text{Span}\{v_1, v_2, v_3\}$. Equivalently, elimination of $[A \mid u]$ creates an inconsistent row.

Exercise 2.8.7: A finite generating set versus its infinite span

(a) The set $\{v_1, v_2, v_3\}$ contains exactly three distinct vectors.

(b) Its column space contains infinitely many vectors: all multiples tv_1 are in it, and $v_1 \neq 0$ makes these distinct for distinct real t.

(c) Yes, $p \in \text{Col}\, A$. Solving $Ac = p$ gives the reduced equations $c_1 + (7/4)c_3 = -9/4$, $c_2 + (5/2)c_3 = -7/2$. Choosing the free variable $c_3 = 0$ yields $p = (-9/4)v_1 - (7/2)v_2$. Its coordinates are $(6, -10, 11)$, as required.

Exercise 2.8.8: Another column-space membership test

A convenient combination with integer coefficients is

$$v_1 + 2v_2 + v_3 = (-2, 0, 6)^T + 2(-2, 3, 3)^T + (0, -5, 5)^T = (-6, 1, 17)^T = p.$$

Thus $p \in \operatorname{Col} A$. One could instead reduce $[A \mid p]$; the displayed coefficients give a direct, independently checkable witness.

Exercise 2.8.9: Column membership is not null membership

With A, p from Exercise 7, compute

$$Ap = \begin{bmatrix} 2(6) - 3(-10) - 4(11) \\ -8(6) + 8(-10) + 6(11) \\ 6(6) - 7(-10) - 7(11) \end{bmatrix} = \begin{bmatrix} -2 \\ -62 \\ 29 \end{bmatrix} \neq 0.$$

Therefore $p \notin \operatorname{Nul} A$, even though Exercise 7 placed p in $\operatorname{Col} A$.

Exercise 2.8.10: A nonzero null vector

For $u = (-5, 5, 3)^T$ and the matrix in Exercise 8,

$$Au = \begin{bmatrix} 10 - 10 + 0 \\ 0 + 15 - 15 \\ -30 + 15 + 15 \end{bmatrix} = 0.$$

Hence $u \in \operatorname{Nul} A$. Every coordinate has been checked; the nonzero input being sent to zero is exactly the definition of null-space membership.

Exercise 2.8.11: Ambient dimensions from the matrix shape

The given matrix has three rows and four columns. An input to A has four entries and an output has three, so $\operatorname{Nul} A \subseteq \mathbb{R}^4$ and $\operatorname{Col} A \subseteq \mathbb{R}^3$. Thus $\boxed{p = 4, q = 3}$. These are ambient dimensions, not necessarily the dimensions of the two subspaces themselves.

Exercise 2.8.12: Five outputs and three inputs

Here A is 5×3. Consequently $\operatorname{Nul} A \subseteq \mathbb{R}^3$ and $\operatorname{Col} A \subseteq \mathbb{R}^5$, giving $\boxed{p = 3, q = 5}$. Count columns for inputs and rows for outputs; these roles should not be reversed.

Exercise 2.8.13: One null vector and one column vector

For the matrix in Exercise 11, use

$$z = (1, -2, 1, 0)^T, \qquad a_1 = (3, -9, 9)^T.$$

Multiplication gives $Az = (3 - 4 + 1, -9 + 8 + 1, 9 - 4 - 5)^T = 0$, so the nonzero z is in $\operatorname{Nul} A$. The nonzero a_1 belongs to $\operatorname{Col} A$ because it is an actual column, namely Ae_1. The two vectors have different numbers of entries, as their respective ambient spaces require.

Exercise 2.8.14: Choose a null vector with integer entries

For the 5×3 matrix in Exercise 12, take $z = (1, -5, 3)^T$. Its row products are $1 - 10 + 9$, $4 - 25 + 21$, $-5 + 5 + 0$, $2 - 35 + 33$, and $3 - 15 + 12$, all zero. Thus $z \neq 0$ belongs to $\operatorname{Nul} A$. A nonzero column-space vector is $a_1 = (1, 4, -5, 2, 3)^T$. It belongs to $\operatorname{Col} A$ by its definition as column 1.

Exercise 2.8.15: A basis of the plane

The two vectors $(4, -2)^T$ and $(16, -3)^T$ have determinant $4(-3) - 16(-2) = 20 \neq 0$. Their 2×2 column matrix is invertible, so they are independent and span $\mathbb{R}^2$. Therefore they form a basis for $\mathbb{R}^2$.

Exercise 2.8.16: A dependent pair cannot be a basis

The second vector $(4, -10)^T$ is -2 times the first $(-2, 5)^T$. Thus the set is dependent and spans only one line. It is not a basis for $\mathbb{R}^2$, which requires two independent directions.

Exercise 2.8.17: Three pivots after a cyclic row reordering

Place the given vectors into columns: $A = \begin{bmatrix} 0 & 5 & 6 \\ 0 & 0 & 3 \\ -2 & 4 & 2 \end{bmatrix}$. Moving row 3 to the top gives the echelon matrix $\begin{bmatrix} -2 & 4 & 2 \\ 0 & 5 & 6 \\ 0 & 0 & 3 \end{bmatrix}$. Its three pivots show independence and spanning, so the given vectors form a basis for $\mathbb{R}^3$.

Exercise 2.8.18: An explicit dependence in three dimensions

The third vector equals twice the first plus the second: $2(1, 1, -3)^T + (3, -1, 2)^T = (5, 1, -4)^T$. Thus $2v_1 + v_2 - v_3 = 0$ is a nontrivial dependence relation. The set is not a basis for $\mathbb{R}^3$.

Exercise 2.8.19: Too few vectors to span three-dimensional space

Only two vectors are supplied. Their 3×2 column matrix has at most two pivots, so it cannot have a pivot in every one of its three rows. They do not span $\mathbb{R}^3$ and therefore do not form a basis for it. Their own span can have a basis of two vectors; that is a different question.

Exercise 2.8.20: Too many vectors for independence in three dimensions

Four vectors in $\mathbb{R}^3$ are necessarily dependent: the associated 3×4 homogeneous system has at least one free variable. Hence these four vectors do not form a basis for $\mathbb{R}^3$, regardless of whether they happen to span the whole space.

Exercise 2.8.21: Subspace definitions and preserved dependence

(a) **False as written.** The closure requirements must say *for every* $u, v \in H$, $u + v \in H$ and *for every* $u \in H$ *and every scalar* c, $cu \in H$. Simply listing that some $u, v, u + v, cu$ are in H is insufficient. For example $H = \{0, e_1, -e_1\}$ has zero and satisfies those listed facts with $u = v = 0$, but $2e_1 \notin H$. With the universal conditional quantifiers restored, the statement is exactly the correct subspace definition.

(b) **True.** $[v_1 \ \cdots \ v_p]x$ is precisely a linear combination of the listed vectors, so its set of possible values is their span.

(c) **False.** There are n unknowns, so the solution vectors have n entries; the subspace is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(d) **True.** The IMT supplies both independence and spanning for the columns of an invertible $n \times n$ matrix.

(e) **True.** If row operations give $B = EA$ with invertible E, then $Ac = 0$ if and only if $Bc = 0$. Thus precisely the same coefficient vectors express dependencies among the corresponding columns.

Exercise 2.8.22: Distinguish spaces from their descriptions

(a) **False.** $H = \{0, e_1\}$ contains zero but not $2e_1$, so it fails scalar closure.

(b) **False.** Pivot columns must be selected from A itself. For example $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ reduces to $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Its pivot column e_1 is not in $\operatorname{Col} A = \operatorname{Span}\{e_2\}$.

(c) **True.** The set contains zero (all coefficients zero), and sums or scalar multiples of linear combinations are linear combinations. These are exactly the three subspace tests.

(d) **False.** Closure requires both added vectors in H. Take $H = \operatorname{Span}\{e_1\} \subset \mathbb{R}^2$, $x = 0$, and $y = e_2$; then $x + y = e_2 \notin H$.

(e) **False.** $\operatorname{Col} A$ is the set of right sides b for which $Ax = b$ has a solution. It is not the set of solution inputs x to one fixed equation; input and output spaces can even have different dimensions.

Exercise 2.8.23: Two pivot columns and two free variables

The given echelon form has pivots in columns 1 and 2. Therefore a column-space basis is

$$\mathcal{C} = \{(4, 6, 3)^T, (5, 5, 4)^T\},$$

taken from the *original* matrix. For $Ax = 0$, the echelon equations are $x_1 + 2x_2 + 6x_3 - 5x_4 = 0$ and $x_2 + 5x_3 - 6x_4 = 0$. Set $x_3 = s$, $x_4 = t$; then $x_2 = -5s + 6t$ and $x_1 = 4s - 7t$. Thus

$$x = s(4, -5, 1, 0)^T + t(-7, 6, 0, 1)^T,$$

and the two displayed coefficient vectors form a basis for $\operatorname{Nul} A$. They span because this is the general solution, and they are independent because their free coordinates x_3, x_4 force $s = t = 0$ in a zero combination.

Exercise 2.8.24: A nonadjacent pair of pivot columns

The pivot columns are 1 and 3, so a basis for $\operatorname{Col} A$ is $\{(3,2,3)^T, (9,7,6)^T\}$. The echelon equations reduce to $x_3 = -2x_4$ and $x_1 - 2x_2 - 6x_4 = 0$. Let $x_2 = s$, $x_4 = t$. Then

$$x = s(2,1,0,0)^T + t(6,0,-2,1)^T.$$

The coefficient vectors form a null-space basis: they describe all solutions and their free-coordinate entries make them independent.

Exercise 2.8.25: Three pivot columns among five inputs

The echelon form has pivot columns 1,2,4. A column-space basis is

$$\{(1,-1,-2,3)^T, (4,2,2,6)^T, (-3,3,5,-5)^T\}.$$

The three equations are $x_1 + 4x_2 + 8x_3 + 5x_5 = 0$, $2x_2 + 5x_3 - x_5 = 0$, and $x_4 + 4x_5 = 0$. Taking $x_3 = s$, $x_5 = t$ gives $x_2 = -5s/2 + t/2$, $x_4 = -4t$, $x_1 = 2s - 7t$. Hence a basis for $\operatorname{Nul} A$ is

$$\{(2,-5/2,1,0,0)^T, (-7,1/2,0,-4,1)^T\}.$$

Substituting either vector into the three echelon equations yields zero; independence follows from their third and fifth coordinates.

Exercise 2.8.26: Another three-pivot example

The pivot positions are in columns 1,2,4 of the original matrix. A column-space basis is

$$\{(3,3,0,6)^T, (-1,1,3,3)^T, (-1,0,-1,-2)^T\}.$$

The echelon equations give $x_4 = 2x_5$, $x_2 = -3x_3 + 2x_5$, and $3x_1 - x_2 - 3x_3 + 6x_5 = 0$, so $x_1 = -4x_5/3$. Thus a null-space basis is

$$\{(0,-3,1,0,0)^T, (-4/3,2,0,2,1)^T\}.$$

The free variables are x_3, x_5; this explains both why there are two basis vectors and why they are independent.

Exercise 2.8.27: A column combination different from every column

Choose $A = I_3$ and $b = (1,1,0)^T$. Then $b = e_1 + e_2 = A(1,1,0)^T \in \operatorname{Col} A$, and $b \neq 0$. It differs from each of e_1, e_2, e_3, the columns of A. Column space contains all combinations, not just the listed columns.

Exercise 2.8.28: Construct an unattainable output

Take $A = \operatorname{diag}(1,1,0)$ and $b = e_3$. Every output Ax has third coordinate zero, whereas b has third coordinate one. Hence $b \notin \operatorname{Col} A$.

Exercise 2.8.29: A nonzero input annihilated by a nonzero matrix

Take $A = \operatorname{diag}(1, 0, 0)$ and $b = e_2$. Both A and b are nonzero, but $Ab = 0$, so $b \in \operatorname{Nul} A$. This does not violate any rule about scalars: nonzero matrices can annihilate nonzero vectors.

Exercise 2.8.30: Independent columns are already a basis for their own span

By definition $\operatorname{Col} A = \operatorname{Span}\{a_1, \ldots, a_p\}$, so the columns span this space. The given assumption supplies independence. These are exactly the two conditions for a basis, hence the columns are a basis for $\operatorname{Col} A$. No claim that the column space equals the whole codomain is needed.

Exercise 2.8.31: A proper column space of a square matrix

Since $\operatorname{Col} F \neq \mathbb{R}^5$, the 5×5 matrix is not onto. The IMT makes it singular, so $\operatorname{Nul} F$ contains a nonzero vector. It is a subspace of $\mathbb{R}^5$ and contains infinitely many vectors, since all real multiples of any nonzero null vector are distinct members. The exact size of a basis is not determined by the premise.

Exercise 2.8.32: Full column space in seven dimensions

$\operatorname{Col} B = \mathbb{R}^7$ says every b is attained. Because B is square, the IMT makes it invertible. Thus for every $b \in \mathbb{R}^7$, $Bx = b$ has exactly one solution, $B^{-1}b$. In particular $\operatorname{Nul} B = \{0\}$; there are no free variables.

Exercise 2.8.33: A trivial null space in six dimensions

The only homogeneous solution is zero, so the square matrix C is invertible by the IMT. Therefore for every $b \in \mathbb{R}^6$ the equation $Cx = b$ has a unique solution $C^{-1}b$, and $\operatorname{Col} C = \mathbb{R}^6$. The conclusion would require qualification for a nonsquare matrix.

Exercise 2.8.34: A column basis of the entire codomain fixes the shape

Spanning $\mathbb{R}^m$ requires a pivot in every row, so there are m pivots. Independence requires every column to be a pivot column, so there are n pivots. Hence $m = n$: A is square. It is then invertible by the IMT. Both basis requirements, rather than spanning alone, forced equality.

Exercise 2.8.35: A nontrivial null space implies a proper column space

The given nonzero null vector makes the 5×5 matrix singular. Therefore its column space is a proper subspace of $\mathbb{R}^5$ by the IMT. Some right sides are unattainable. Every attainable right side has infinitely many preimages: if $Bx = b$ and $Bz = 0$ with $z \neq 0$, then $B(x + tz) = b$ for every real t.

Exercise 2.8.36: Independent columns of a tall matrix

The four columns are independent, so $Cx = 0$ has only $x = 0$. Hence $\boxed{\operatorname{Nul} C = \{0\} \subseteq \mathbb{R}^4}$. Although there are six output coordinates, inputs and null vectors have four coordinates. The independent columns do not span $\mathbb{R}^6$, so not every right side is attained, despite uniqueness whenever one is attained.

Exercise 2.8.37: An exact computer-assisted pair of bases

Exact row reduction of the supplied matrix gives

$$R = \begin{bmatrix} 1 & 0 & 5/2 & -9/2 & 7/2 \\ 0 & 1 & 3/2 & -5/2 & 3/2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1 and 2, so $\{(3,-7,-5,3)^T, (-5,9,7,-7)^T\}$ is a column-space basis. For the null space, solve $x_1 = -5x_3/2 + 9x_4/2 - 7x_5/2$ and $x_2 = -3x_3/2 + 5x_4/2 - 3x_5/2$. Taking each free variable in turn equal to 2 gives the integer basis

$$\{(-5,-3,2,0,0)^T, (9,5,0,2,0)^T, (-7,-3,0,0,2)^T\}.$$

Scaling basis vectors by 2 does not change their span or independence. These were checked by exact multiplication by the original A. A reproducible computation is `R,pivots=Matrix(A).rref()` in SymPy, followed by `Matrix(A).nullspace()`; the latter may return scalar multiples of the chosen vectors.

Exercise 2.8.38: Four pivot columns and one free coordinate

Exact row reduction gives

$$R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Thus original columns 1,2,4,5 form a column-space basis. Explicitly they are the columns of

$$C = \begin{bmatrix} 5 & 3 & -6 & -8 \\ 4 & 1 & -8 & -7 \\ 5 & 1 & 5 & 19 \\ -7 & -5 & 8 & 5 \end{bmatrix}.$$

For $Rx = 0$, $x_1 = -x_3$, $x_2 = x_3$, and $x_4 = x_5 = 0$. Hence $\{(-1,1,1,0,0)^T\}$ is a basis for $\operatorname{Nul} A$. Multiplication by the original matrix checks the null vector, and the four pivot positions establish independence of the selected original columns.